



Product Requirements Document

A comprehensive specification for the Turf Now college sports turf booking and approval system

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This PRD defines the complete functional, technical, and operational requirements for Turf Now — a secure, multi-role turf booking platform enabling CMR University students to reserve sports turf slots with a structured two-level approval workflow involving the Secretary of Sports and the Director.

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1. Executive Summary & Product Vision

Turf Now is a secure, role-based sports turf booking platform purpose-built for CMR University. It replaces informal, chaotic turf reservation practices — WhatsApp messages, paper sign-up sheets, and verbal commitments — with a structured digital workflow that ensures fair access, accountability, and transparent resource allocation.

The platform introduces a **two-level approval hierarchy**: after a student submits a booking request, it must be approved first by the **Secretary of Sports** and then by the **Director** before the slot is confirmed. This governance model ensures institutional oversight while automating the administrative burden of manual coordination.

Product Vision Statement

"To become the single source of truth for all sports turf reservations at CMR University, eliminating booking conflicts, ensuring equitable access, and providing real-time visibility into turf utilization for students and administration alike."

Key Differentiators

Two-Level Approval: Unlike generic booking apps, Turf Now enforces institutional oversight through Secretary to Director approval chains.

Academic Identity Integration: Registration tied to university credentials (Register Number, Department, Year, Section) — no anonymous bookings.

Waitlist with Auto-Promotion: If a booking is cancelled or rejected, the next student in the waitlist is automatically notified and promoted.

Maintenance Windows: Admins can block turf for maintenance with automatic notification to affected students.

Audit Trail: Every booking action (create, approve, reject, cancel) is logged with timestamp and actor for institutional accountability.

2. Problem Statement & Market Context

2.1 Current State Pain Points

Booking Chaos: Students currently reserve turf through informal channels — WhatsApp groups, physical sign-up boards, or verbal agreements with the Secretary. This leads to double-bookings, forgotten reservations, and disputes over who 'called it first'.

Lack of Transparency: There is no centralized view of who has booked the turf when. Students arrive at the turf only to find it occupied, wasting time and creating friction.

Approval Bottleneck: The Secretary of Sports manually reviews every request via messages or in-person. This is time-consuming, error-prone, and creates delays — especially during peak tournament seasons.

No Accountability: Without a digital record, students who no-show or damage equipment cannot be held accountable. There is no evidence trail for disciplinary action.

Maintenance Blind Spots: Turf maintenance schedules are communicated ad-hoc. Students show up to find the turf closed, with no prior notification.

Inequitable Access: Popular slots (evenings, weekends) are often captured by the same groups repeatedly, while other students are left without access.

2.2 Market Context

While commercial turf booking platforms exist (Playo, Sporloc, KheloMore), they are designed for pay-per-use commercial venues and lack the institutional governance layer required by universities. Enterprise resource management systems (Oracle, SAP) are over-engineered and cost-prohibitive for a single university turf. Turf Now occupies the niche of **institutional sports facility management with embedded approval workflows** — a segment largely underserved by existing solutions.

3. User Personas & Roles

The platform serves four distinct user roles, each with specific permissions, workflows, and interface requirements:

Student

Primary end-user. Books turf slots, views own bookings, manages profile, contacts Secretary for queries. Cannot book without approval.

Secretary of Sports

First-level approver. Reviews all student bookings, approves/rejects with comments, manages turf rules, handles student queries.

Director

Second-level approver. Final authority on bookings. Views analytics, makes announcements, overrides Secretary decisions if needed.

System Admin

Technical administrator. Manages user accounts, configures maintenance windows, views audit logs, exports data.

Role Permission Matrix

Capability	Student	Secretary	Director	Admin
Register / Login	Yes	Yes	Yes	Yes
Book Turf Slot	Yes	Yes	Yes	No
View Own Bookings	Yes	Yes	Yes	No
Cancel Own Booking	Yes	Yes	Yes	No
Approve/Reject Bookings	No	Yes	Yes	No
View All Bookings	No	Yes	Yes	Yes
Manage Turf Rules	No	Yes	Yes	No
Post Announcements	No	No	Yes	Yes
View Analytics	No	No	Yes	Yes
Manage Maintenance	No	No	No	Yes
Export Data	No	No	No	Yes
View Audit Logs	No	No	No	Yes
Manage Users	No	No	No	Yes

Table 1: Role-based permission matrix

4. Functional Requirements

4.1 Authentication & Onboarding (FR-AUTH-001 to FR-AUTH-012)

FR-AUTH-001: Students must register using university email domain (@cmr.edu.in) to ensure institutional identity.

FR-AUTH-002: Registration fields: Full Name, Email, Register Number, Building, Department, Year, Section, Phone Number, Password, Confirm Password.

FR-AUTH-003: Password must be minimum 8 characters with at least one uppercase, one lowercase, one number, and one special character.

FR-AUTH-004: System must verify email via OTP before account activation.

FR-AUTH-005: Register Number must be unique across the system; duplicate registration attempts must be rejected.

FR-AUTH-006: Login requires Email + Password. Failed attempts are rate-limited (5 attempts per 15 minutes).

FR-AUTH-007: Forgot Password flow: email verification to secure token link to password reset form.

FR-AUTH-008: Session tokens (JWT) expire after 24 hours of inactivity.

FR-AUTH-009: Onboarding screen displays turf rules, booking policy, and help center links for first-time logins.

FR-AUTH-010: Secretary and Director accounts are pre-provisioned by System Admin; they cannot self-register.

FR-AUTH-011: All authentication events (login, logout, password reset) are logged in the Audit Log.

FR-AUTH-012: Profile page allows students to update Phone Number, Building, and Section. Email and Register Number are immutable.

4.2 Student Booking Flow (FR-BOOK-001 to FR-BOOK-015)

FR-BOOK-001: Student navigates to 'Book Turf' and selects a date from an interactive calendar. Past dates are disabled.

FR-BOOK-002: System displays available time slots in 1-hour increments (e.g., 6:00 AM, 7:00 AM, ..., 10:00 PM).

FR-BOOK-003: Slots already booked (pending or confirmed) are visually marked as unavailable.

FR-BOOK-004: Slots during maintenance windows are disabled with a 'Maintenance' tooltip.

FR-BOOK-005: Student can book maximum 2 slots per day and maximum 5 slots per week.

FR-BOOK-006: Each booking requires: Date, Time Slot, Purpose (Practice / Match / Tournament / Other), and Number of Players (1-22).

FR-BOOK-007: Upon submission, booking status = 'Pending Secretary Approval'. Student receives confirmation notification.

FR-BOOK-008: Student can view all their bookings on 'My Bookings' page with status badges: Pending, Approved by Secretary, Approved by Director, Confirmed, Rejected, Cancelled.

FR-BOOK-009: Student can cancel a booking at any time before it is 'Confirmed'. Cancellation reason is optional but encouraged.

FR-BOOK-010: Cancelled slots are immediately released back to the pool and the first waitlisted student is auto-promoted.

FR-BOOK-011: Student can join a waitlist for fully booked slots. Waitlist position is displayed.

FR-BOOK-012: Student receives notification when waitlisted slot becomes available (auto-promotion or cancellation).

FR-BOOK-013: Booking modification is allowed only before Secretary approval. After approval, student must cancel and re-book.

FR-BOOK-014: System enforces a 30-minute buffer between consecutive bookings by different groups to allow turf turnover.

FR-BOOK-015: Students can download a booking confirmation PDF with QR code for entry verification at the turf gate.

4.3 Approval Workflow Engine (FR-APPR-001 to FR-APPR-010)

FR-APPR-001: All bookings follow a strict sequential approval: Student submits to Secretary reviews to Director reviews to Confirmed.

FR-APPR-002: Secretary receives a notification (in-app + email) for every new booking request.

FR-APPR-003: Secretary can Approve, Reject, or Request More Info on any pending booking.

FR-APPR-004: If Secretary approves, booking status changes to 'Pending Director Approval' and Director is notified.

FR-APPR-005: If Secretary rejects, booking status changes to 'Rejected by Secretary' with mandatory rejection reason. Student is notified.

FR-APPR-006: Director receives a notification for every Secretary-approved booking.

FR-APPR-007: Director can Approve or Reject. Approval changes status to 'Confirmed'. Rejection changes status to 'Rejected by Director' with reason.

FR-APPR-008: Director can override a Secretary rejection if deemed necessary, with an override reason logged in the audit trail.

FR-APPR-009: Approval actions must include a timestamp, actor ID, and optional comment.

FR-APPR-010: Bookings pending approval for more than 48 hours are auto-escalated to the Director with a flag.

4.4 Admin & Secretary Dashboard (FR-ADMIN-001 to FR-ADMIN-012)

FR-ADMIN-001: Admin Dashboard displays: Total Bookings (Today/Week/Month), Pending Approvals, Confirmed Bookings, No-show Rate, Turf Utilization %.

FR-ADMIN-002: Admin Calendar shows all bookings color-coded by status: Pending (Yellow), Confirmed (Green), Cancelled (Red), Maintenance (Gray).

FR-ADMIN-003: Approvals page lists all pending bookings with filters: By Date, By Status, By Student, By Department.

FR-ADMIN-004: Secretary can bulk-approve up to 10 bookings at once for efficiency during high-volume periods.

FR-ADMIN-005: Manage Bookings page allows admins to force-cancel any booking with a reason (e.g., emergency closure).

FR-ADMIN-006: Announcements page allows Director to post broadcast messages visible to all students on login.

FR-ADMIN-007: Turf Rules page is editable by Secretary. Changes are versioned and students must acknowledge updates.

FR-ADMIN-008: Booking Policy page defines: max slots per day/week, cancellation window (e.g., 2 hours before), no-show penalties.

FR-ADMIN-009: Maintenance page allows blocking specific dates/times with a reason. Affected students are auto-notified.

FR-ADMIN-010: Analytics page shows: Peak booking hours, Department-wise usage, Monthly trend charts, Waitlist conversion rate.

FR-ADMIN-011: Audit Logs page shows chronological list of all system actions with: Timestamp, Actor, Action, Target, Before/After state.

FR-ADMIN-012: Admin Settings page allows configuring: notification preferences, approval timeout, auto-escalation rules, and system branding.

4.5 Notifications & Communication (FR-NOTIF-001 to FR-NOTIF-008)

FR-NOTIF-001: In-app notification bell displays unread count and categorized notifications: Booking, Approval, System, Maintenance.

FR-NOTIF-002: Email notifications sent for: Registration confirmation, Booking submission, Approval/Rejection, Waitlist promotion, Maintenance alert.

FR-NOTIF-003: SMS notifications sent for: Booking reminder (2 hours before), Urgent maintenance cancellation, Waitlist promotion.

FR-NOTIF-004: Push notifications (if PWA enabled) for real-time approval updates.

FR-NOTIF-005: Students can configure notification preferences: Email ON/OFF, SMS ON/OFF, Push ON/OFF.

FR-NOTIF-006: Contact Secretary page allows students to send direct messages with category: Booking Query, Complaint, General.

FR-NOTIF-007: Secretary receives contact messages in a dedicated inbox with read receipts and reply capability.

FR-NOTIF-008: Help Center provides FAQ, booking guide video, and contact information for technical support.

4.6 Waitlist & Maintenance (FR-WAIT-001 to FR-WAIT-006)

FR-WAIT-001: When a slot is fully booked, students can join the waitlist. Maximum 5 students per slot waitlist.

FR-WAIT-002: Waitlist is ordered by timestamp (FIFO). Students see their position (e.g., 'You are #2 in line').

FR-WAIT-003: When a confirmed booking is cancelled, the first waitlisted student is auto-promoted to 'Pending Secretary Approval'.

FR-WAIT-004: Promoted waitlist students have 2 hours to confirm interest; otherwise, the slot moves to the next in line.

FR-WAIT-005: Maintenance windows can be scheduled as recurring (e.g., every Sunday 6-8 AM) or one-time.

FR-WAIT-006: Students with bookings affected by maintenance are notified 24 hours in advance with alternative slot suggestions.

5. Non-Functional Requirements

Category	Requirement	Target / Constraint
Performance	Page load time	< 2 seconds on 3G network
Performance	Booking submission latency	< 500ms
Performance	Concurrent users	Support 200 simultaneous users
Availability	Uptime SLA	99.5% excluding scheduled maintenance
Availability	Maintenance windows	Sundays 2-4 AM IST only
Security	Password storage	bcrypt with salt rounds >= 12
Security	Session management	JWT with 24h expiry, HttpOnly cookies
Security	Data transmission	TLS 1.3 minimum
Security	Rate limiting	100 requests/minute per IP
Scalability	Database records	Handle 50,000 bookings without performance degradation
Usability	Mobile responsiveness	Fully functional on 320px width screens
Usability	Accessibility	WCAG 2.1 AA compliance
Reliability	Data backup	Daily automated backups retained for 30 days
Reliability	Audit log retention	7 years (institutional compliance)

Table 2: Non-functional requirements specification

6. Data Model & Schema

The following entity-relationship model defines the core data structures for Turf Now:

6.1 Core Entities

User

id (PK), email, password_hash, full_name, register_number, building, department, year, section, phone, role (student/secretary/director/admin), is_active, created_at, updated_at

Booking

id (PK), user_id (FK), date, start_time, end_time, purpose, num_players, status (pending_secretary/pending_director/confirmed/rejected_secretary/rejected_director/cancelled), secretary_comment, director_comment, created_at, updated_at

Approval

id (PK), booking_id (FK), approver_id (FK), approver_role, action (approve/reject/override), comment, created_at

Waitlist

id (PK), booking_id (FK), user_id (FK), position, status (waiting/promoted/expired), created_at

Maintenance

id (PK), start_datetime, end_datetime, reason, is_recurring, recurrence_pattern, created_by (FK), created_at

Notification

id (PK), user_id (FK), type (booking/approval/system/maintenance), title, message, is_read, channel (in_app/email/sms), created_at

AuditLog

id (PK), actor_id (FK), action, target_type, target_id, before_state, after_state, ip_address, user_agent, created_at

Announcement

id (PK), title, content, posted_by (FK), is_active, created_at, updated_at

TurfRule

id (PK), title, content, version, effective_date, created_by (FK), created_at

6.2 Relationships

- User 1:N Booking (one student can have many bookings)
- Booking 1:N Approval (one booking can have multiple approval actions)
- Booking 1:N Waitlist (one booking slot can have multiple waitlisted students)

- User 1:N Notification (one user receives many notifications)
- User 1:N AuditLog (one actor performs many actions)
- Booking N:1 Maintenance (maintenance blocks many potential bookings)

7. API Specifications

RESTful API endpoints organized by resource. All endpoints require Bearer token authentication except /auth/*.

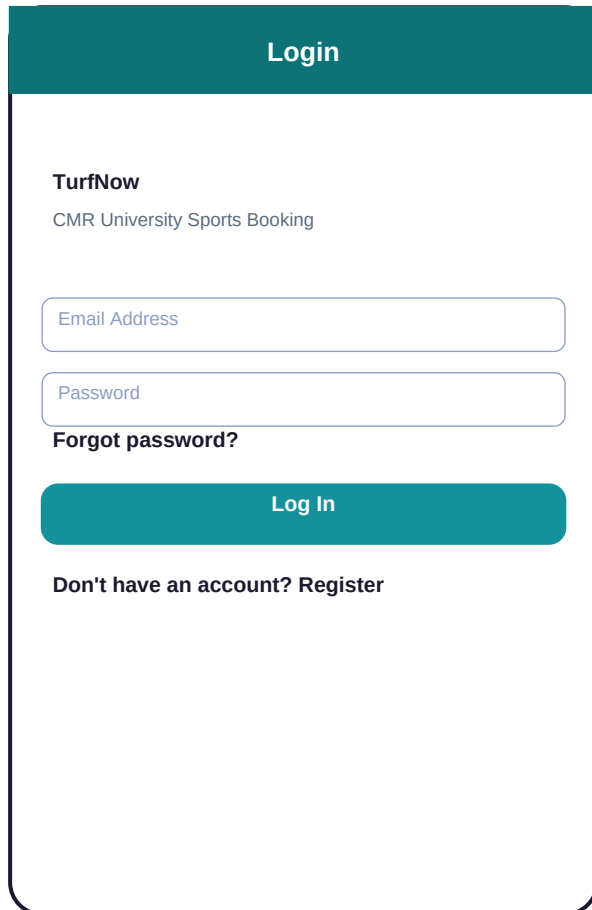
Method	Endpoint	Description	Auth Required
POST	/api/auth/register	Student registration with validation	No
POST	/api/auth/login	Email + password login, returns JWT	No
POST	/api/auth/forgot-password	Initiates password reset flow	No
POST	/api/auth/reset-password	Completes password reset with token	No
GET	/api/auth/me	Returns current user profile	Yes
PUT	/api/auth/profile	Updates mutable profile fields	Yes
POST	/api/bookings	Creates a new booking request	Yes (Student)
GET	/api/bookings/my	Lists current user bookings	Yes
GET	/api/bookings/:id	Retrieves booking details	Yes
PUT	/api/bookings/:id/cancel	Cancels own booking	Yes (Owner)
GET	/api/bookings/pending	Lists all pending approvals	Yes (Sec/Dir)
POST	/api/bookings/:id/approve	Approves a booking	Yes (Sec/Dir)
POST	/api/bookings/:id/reject	Rejects a booking with reason	Yes (Sec/Dir)
GET	/api/slots/available	Returns available slots for date	Yes
POST	/api/waitlist/join	Joins waitlist for a slot	Yes (Student)
GET	/api/waitlist/status	Returns user waitlist positions	Yes
GET	/api/admin/analytics	Returns dashboard metrics	Yes (Dir/Admin)
GET	/api/admin/audit-logs	Returns paginated audit logs	Yes (Admin)
POST	/api/admin/maintenance	Creates maintenance window	Yes (Admin)
GET	/api/notifications	Lists user notifications	Yes
PUT	/api/notifications/:id/read	Marks notification as read	Yes
GET	/api/announcements	Lists active announcements	Yes
POST	/api/contact/secretary	Sends message to Secretary	Yes (Student)

Table 3: Core API endpoint specification

8. User Interface / Wireframes

The following wireframes illustrate the key screens of the Turf Now application. Mobile-first design is prioritized for students; admin interfaces are desktop-optimized.

8.1 Student Mobile Screens



The wireframe shows a mobile login screen. At the top is a dark teal header with the word "Login" in white. Below the header, the text "TurfNow" is followed by "CMR University Sports Booking". There are two input fields: "Email Address" and "Password". Below the password field is a link "Forgot password?". A large teal button labeled "Log In" is centered below the link. At the bottom, the text "Don't have an account? Register" is displayed.

Login

TurfNow
CMR University Sports Booking

Email Address

Password

[Forgot password?](#)

Log In

Don't have an account? [Register](#)

Figure 1: Student Login Screen

Book Turf

Select Date

July 2026

S	M	T	W	T	F	S
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15	16	17	18	19	20	21

Available Slots — Wed, July 15

6:00 AM

7:00 AM

8:00 AM

5:00 PM

6:00 PM

7:00 PM

Continue to Details ->

Figure 2: Book Turf — Calendar & Slot Selection

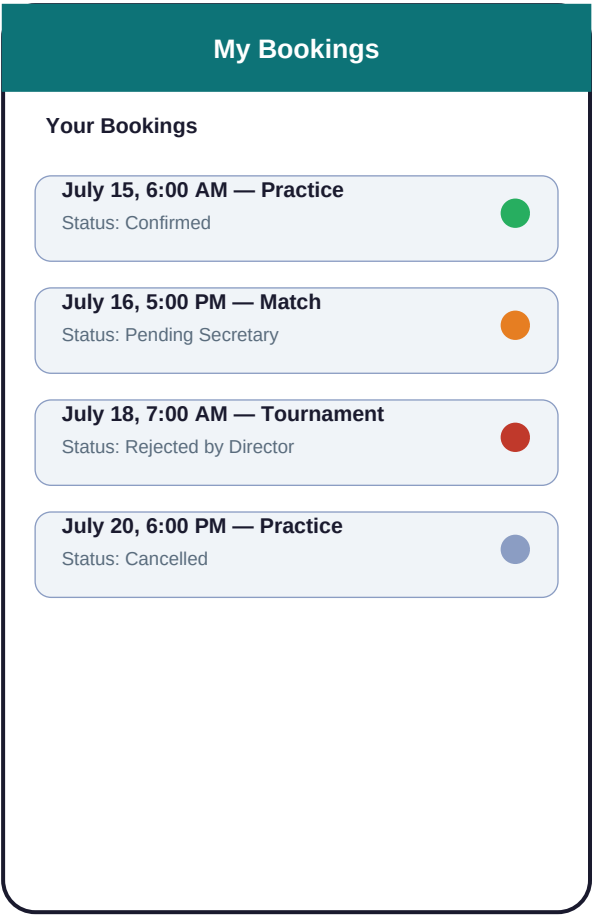


Figure 3: My Bookings — Status Tracking

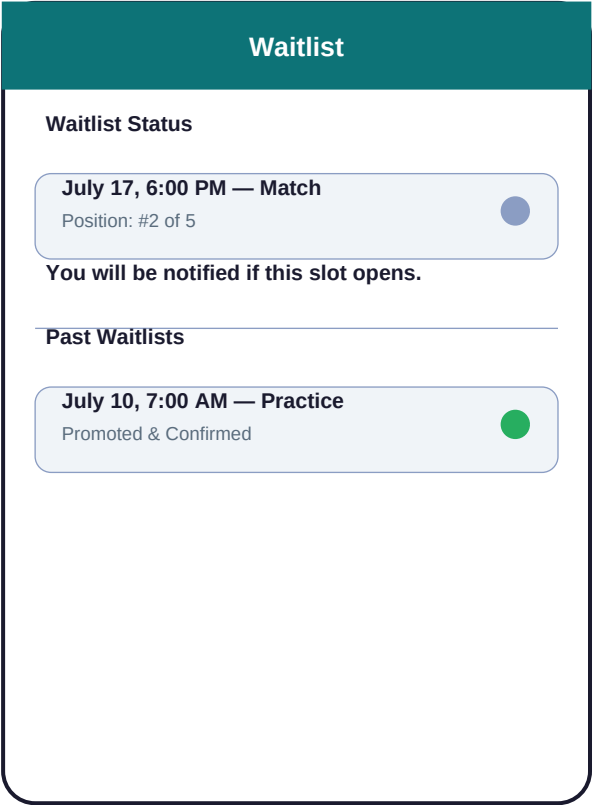


Figure 4: Waitlist Status — Position Tracking

8.2 Admin Desktop Screens

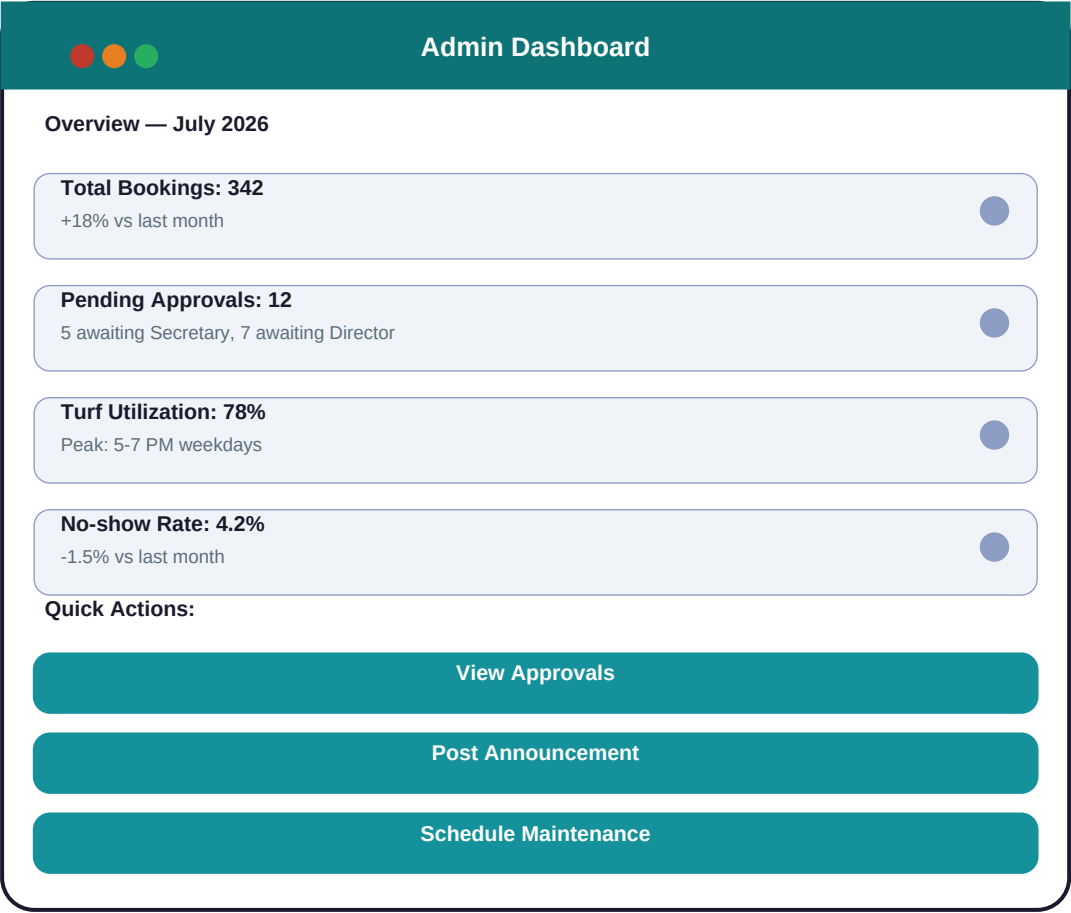


Figure 5: Admin Dashboard — Key Metrics & Actions

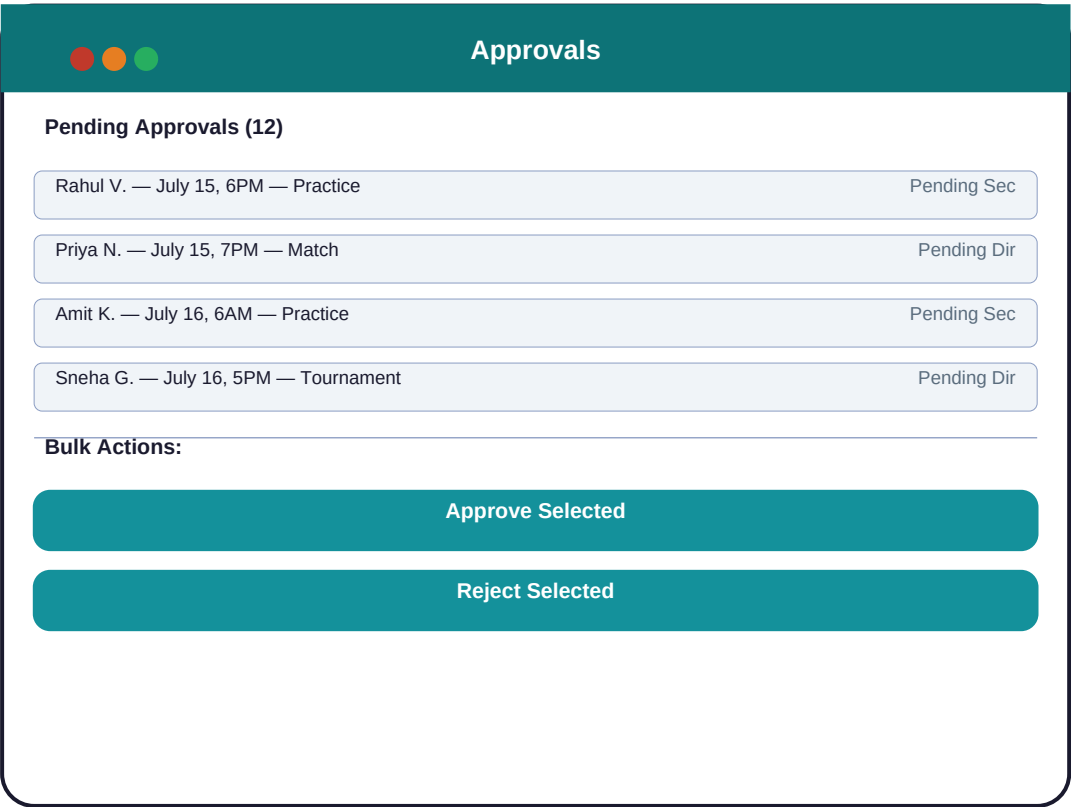


Figure 6: Approvals Queue — Filtered List with Bulk Actions

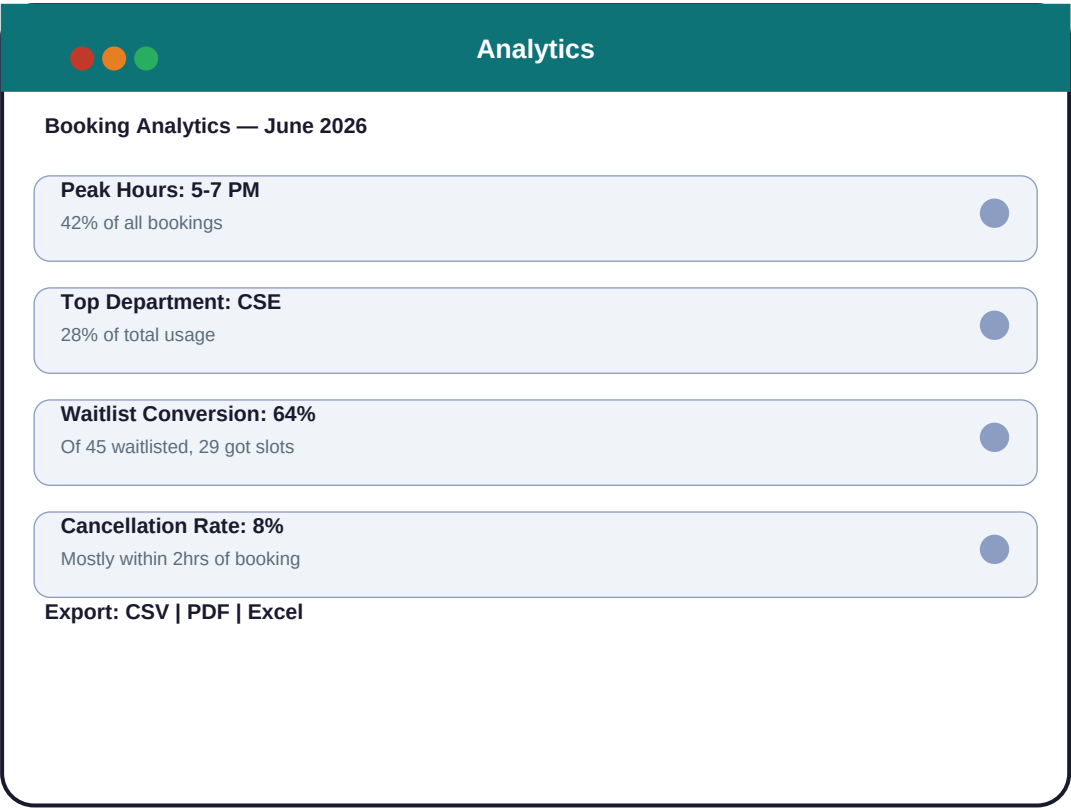


Figure 7: Analytics Dashboard — Usage Insights

9. State Machine & Booking Lifecycle

The booking lifecycle is governed by a finite state machine with 7 states and 10 transitions. Understanding this flow is critical for debugging, testing, and user communication.

9.1 State Definitions

State	Code	Description	Actor Can Modify
Draft	DRAFT	Booking form being filled; not yet submitted	Student
Pending Secretary	PEND_SEC	Submitted; awaiting Secretary review	Secretary
Pending Director	PEND_DIR	Secretary approved; awaiting Director review	Director
Confirmed	CONFIRMED	Director approved; slot reserved	Admin (force-cancel)
Rejected by Secretary	REJ_SEC	Secretary rejected with reason	None (terminal)
Rejected by Director	REJ_DIR	Director rejected with reason	None (terminal)
Cancelled	CANCELLED	Student cancelled or admin force-cancelled	None (terminal)

Table 4: Booking state definitions

9.2 Transition Rules

From State	Action	To State	Actor	Trigger Condition
DRAFT	Submit	PEND_SEC	Student	All required fields valid
PEND_SEC	Approve	PEND_DIR	Secretary	No conflicts, within policy
PEND_SEC	Reject	REJ_SEC	Secretary	Any reason provided
PEND_DIR	Approve	CONFIRMED	Director	Final authorization
PEND_DIR	Reject	REJ_DIR	Director	Any reason provided
PEND_DIR	Override	CONFIRMED	Director	Overrules Secretary rejection
PEND_SEC	Cancel	CANCELLED	Student	Before Secretary action
PEND_DIR	Cancel	CANCELLED	Student	Before Director action
CONFIRMED	Force Cancel	CANCELLED	Admin	Emergency/maintenance
CONFIRMED	No-show	CANCELLED	System	Auto after 15 min grace

Table 5: State transition matrix

9.3 State Diagram (Textual Representation)

```
DRAFT --[Student submits]--> PEND_SEC --[Secretary approves]--> PEND_DIR --[Director approves]--> CONFIRMED
PEND_SEC --[Secretary rejects]--> REJ_SEC
PEND_DIR --[Director rejects]--> REJ_DIR
```

CONFIRMED --[Student cancels]--> CANCELLED

CONFIRMED --[Admin force-cancel]--> CANCELLED

10. Security & Compliance

Authentication Security: JWT tokens with 24-hour expiry. Refresh tokens with 7-day expiry. Passwords hashed with bcrypt (salt rounds 12). Brute-force protection: 5 failed attempts trigger 15-minute lockout.

Authorization: Role-based access control (RBAC) with middleware enforcement on every API endpoint. Students can only access their own bookings. Secretary can only act on pending bookings, not admin settings.

Data Protection: All PII (email, phone, register number) encrypted at rest using AES-256. TLS 1.3 for all data in transit. No plaintext password storage.

Input Sanitization: All user inputs validated via Zod schemas. SQL injection prevention via parameterized queries (ORM). XSS prevention via output encoding.

Audit & Accountability: Every state-changing action is logged: actor ID, timestamp, IP address, user agent, before/after state. Audit logs are append-only and exportable.

Compliance Framework: While Turf Now handles student data (not patient data), it should align with: India's IT Act 2000 (data protection), CMR University data governance policies, and GDPR principles if international students are served.

Incident Response: Security incidents (unauthorized access, data breach) must be reported to the Director within 24 hours. Automated alerts for suspicious login patterns (impossible travel, multiple IPs).

11. Technology Stack

Turf Now is built on the Base44 no-code/low-code platform, which generates a full-stack application from visual design and logic configuration. The following stack is inferred from the platform's typical output and the application's functional requirements.

11.1 Frontend

Framework: [React 18 + TypeScript](#) — Component-based SPA with strong typing.

Styling: [Tailwind CSS](#) — Utility-first responsive design.

State: [React Context + useReducer](#) — Sufficient for app-scale state management.

Routing: [React Router v6](#) — Protected routes with role-based guards.

UI Library: [shadcn/ui + Radix](#) — Accessible, themeable components.

Charts: [Recharts](#) — Analytics dashboard visualizations.

Calendar: [react-calendar](#) or [date-fns](#) — Interactive booking calendar.

PWA: [Workbox](#) — Service worker for offline notifications page.

11.2 Backend

Runtime: [Node.js 20 LTS](#) — Stable, high-performance JavaScript runtime.

Framework: [Express.js](#) — Minimal REST API framework.

Database: [PostgreSQL](#) — Relational database with ACID compliance for booking integrity.

ORM: [Prisma](#) — Type-safe database access with migration support.

Auth: [JWT + bcrypt](#) — Stateless authentication with secure password hashing.

Email: [Nodemailer](#) / [SendGrid](#) — Transactional email delivery.

SMS: [Twilio](#) / [MSG91](#) — OTP and reminder SMS for Indian numbers.

Queue: [Bull](#) + [Redis](#) — Background job processing for notifications and waitlist promotions.

11.3 Infrastructure

Hosting: [Base44 Cloud](#) / [Vercel](#) — Serverless deployment with automatic scaling.

Database Hosting: [Supabase](#) / [Railway](#) — Managed PostgreSQL with automated backups.

CDN: [Cloudflare](#) — Global edge caching for static assets.

Storage: [AWS S3](#) / [Cloudflare R2](#) — Profile photos, booking PDFs, audit log exports.

Monitoring: [Sentry](#) + [UptimeRobot](#) — Error tracking and uptime monitoring.

12. Trade-offs & Architectural Decisions

Decision	Chosen Approach	Alternative	Rationale
Platform	Base44 (No-code)	Custom React + Node.js	Spiced: MVP in days vs weeks. Lower cost. Exportable code.
Approval Flow	Sequential (Sec to Dir)	Parallel (both simultaneous)	Ensures Secretary filters noise before Director. Reduces Director workload.
Waitlist	Auto-promotion (FIFO)	Manual Secretary assignment	Fairness and automation. Reduces admin overhead.
Notifications	Email + SMS + In-app	Email only	Students check SMS more frequently. Critical for time-sensitive approvals.
Database	PostgreSQL	Relational MongoDB	Main table. ACID compliance prevents double-booking race conditions.
Auth	JWT (stateless)	Session-based (stateful)	Scales better with serverless. Easier to implement across micro-frontends.
Mobile	Responsive Web (PWA)	Native iOS/Android	Single codebase. No app store approval delays. Sufficient for university use case.
Maintenance	Scheduled windows	On-demand blocking	Predictable for students. Allows advance planning.

Table 6: Key architectural decisions and trade-offs

13. Success Metrics & KPIs

The following metrics will be tracked to measure Turf Now's impact and guide product iterations:

Category	Metric	Target	Measurement Method
Adoption	Monthly Active Students	> 60% of enrolled students	Unique logins per month
Adoption	Booking Conversion Rate	> 75%	Bookings / Login sessions
Efficiency	Avg Approval Time	< 4 hours	Timestamp: submit to confirm
Efficiency	Secretary Time Saved	> 10 hrs/week	Pre-post deployment comparison
Satisfaction	Student NPS	> 50	Monthly survey
Satisfaction	App Store Rating	> 4.2/5	PWA install feedback
Reliability	Uptime	> 99.5%	UptimeRobot monitoring
Reliability	Booking Conflict Rate	< 0.5%	Manual audit of double-bookings
Engagement	Repeat Booking Rate	> 40%	Students with 2+ bookings/month
Engagement	Waitlist Utilization	> 50%	Promoted waitlist / Total waitlist
Compliance	Audit Log Coverage	100%	All state changes logged
Compliance	Data Backup Success	100%	Daily backup verification

Table 7: Success metrics and KPIs

14. Roadmap & Future Enhancements

Phase	Timeline	Feature	Impact
Phase 1	Current	Core booking + 2-level approval + notifications	MVP operational
Phase 2	Month 1-2	Waitlist auto-promotion + maintenance windows	Reduce manual admin by 40%
Phase 3	Month 3-4	Analytics dashboard + export + audit logs	Data-driven decisions
Phase 4	Month 5-6	Multi-turf support (cricket, basketball, volleyball)	Expand beyond football
Phase 5	Month 7-9	Equipment booking (balls, cones, bibs)	Complete sports ecosystem
Phase 6	Month 10-12	Tournament module + bracket generation	Support intra-college events
Phase 7	Year 2	Mobile native app (Flutter)	Better performance & push notifications
Phase 8	Year 2+	Inter-college booking + payment integration	Revenue generation + expansion

Table 8: Product roadmap

15. Appendix

A. Glossary of Terms

- Booking:** A reservation request for a specific turf slot on a specific date and time.
- Slot:** A 1-hour time interval during which the turf can be reserved (e.g., 6:00 AM - 7:00 AM).
- Approval:** An authorized action (Approve/Reject) taken by the Secretary or Director on a pending booking.
- Waitlist:** A queue of students who want a slot that is currently fully booked.
- Maintenance Window:** A scheduled period during which the turf is unavailable for booking due to upkeep.
- No-show:** A confirmed booking where the student does not arrive within 15 minutes of the slot start time.
- Audit Log:** An immutable record of all system actions for accountability and compliance.
- PWA:** Progressive Web App — a web application that can be installed on mobile devices like a native app.

B. Revision History

Version	Date	Author	Changes
0.1	June 2026	Product Team	Initial draft
0.2	July 2026	Product Team	Added approval workflow, wireframes, API specs
1.0	July 2026	Product Team	Final PRD release

Table 9: Document revision history

C. References

- Base44 Platform Documentation — <https://base44.com/docs>
- CMR University Sports Department Guidelines — Internal Document
- WCAG 2.1 Accessibility Guidelines — <https://www.w3.org/WAI/WCAG21/quickref/>
- India IT Act 2000 (Amended 2008) — Data Protection Provisions
- OWASP Top 10 2021 — Web Application Security Standards